

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE ON POLYMERIZATION, MOLECULAR-WEIGHT CONTROL, AND LATEX WORKUP

1. ABSTRACT

This report synthesizes patent evidence relevant to low acrylonitrile nitrile butadiene rubber (NBR), defined here as NBR compositions and processes in which acrylonitrile is present at the lower end of the conventional NBR range or is otherwise tuned to balance oil resistance, low-temperature flexibility, processability, and cure behavior. The selected patent landscape spans classical emulsion-polymerized NBR, solution-polymerized nitrile rubbers, and more recent latex-processing and post-polymerization modification approaches. Across the set, the dominant synthesis route remains free-radical copolymerization of a conjugated diene with acrylonitrile, usually butadiene/acrylonitrile, with optional third monomers in some formulations. Older disclosures emphasize molecular-weight control using chain-transfer agents such as cycloaliphatic mercaptans or alkylthiols, while later patents broaden the process window to include organic-solution polymerization, in-situ emulsifier formation, and post-polymerization maturation or coagulation control.

The strongest NBR-specific process teachings are found in patents focused on monomer feed strategy and molecular-weight regulation. One family uses staged addition of acrylonitrile and alkylthiol to suppress high-molecular-weight fractions and improve processability while retaining vulcanizability. Another family discloses cycloaliphatic mercaptans as unusually effective chain regulators across solution, emulsion, and suspension routes, with broad monomer and initiator compatibility. A newer route moves NBR synthesis into polar organic solvents, using free-radical initiators and a wide range of molecular-weight modifiers, with optional hydrogenation downstream. In parallel, latex-workup patents show that coagulation and washing conditions strongly affect residual salt/metal content and color stability, which is especially relevant for low-salt, color-stable NBR grades. More recent latex patents also target downstream performance: one uses thiocarbonyl-functional post-treatment to improve mechanical properties of dip-molded articles, and another uses controlled emulsion polymerization with in-situ emulsifier formation, though not NBR-specific in the extracted passages.

Overall, the evidence indicates that low-ACN NBR synthesis is best approached by emulsion polymerization or solution polymerization with careful monomer staging, moderate-to-low acrylonitrile content within the disclosed NBR range, and deliberate molecular-weight control using mercaptan-type modifiers. For industrially relevant low-ACN NBR, the most consistently supported

parameters are staged monomer feed, free-radical initiation, chain-transfer regulation, and coagulation/washing conditions optimized to minimize residual salts and preserve color and processing reliability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: US5807941A
- Patent Title: Unsaturated nitrile-conjugated diene copolymer process for producing same and ...
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 1998
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: EP2611837B1
- Patent Title: Method for producing nitrile rubbers in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 4

- Patent Number: US7829624B2
- Patent Title: One-pot synthesis of nanoparticles and liquid polymer for rubber applications
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 5

- Patent Number: US3862274A
- Patent Title: Preparation of nitrile polymers with cycloaliphatic mercaptans as ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1975
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: US11248078B2
- Patent Title: Color-stable nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 7

- Patent Number: US10336853B2
- Patent Title: Polymer, process and composition
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 8

- Patent Number: US9711785B2
- Patent Title: Negative electrode slurry composition, lithium ion secondary battery negative ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 9

- Patent Number: US20080255314A1
- Patent Title: Dip-forming latex, dip-forming composition and dip-formed article
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2008
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 10

- Patent Number: EP3555198B1
- Patent Title: Curable polymer latex compositions for the manufacture of rubber articles
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US5807941A and US3862274A both address NBR-type copolymerization with chain-transfer control, but US5807941A is more specific to acrylonitrile/butadiene and uses an alkylthiol modifier added partly before polymerization and partly during polymerization, whereas US3862274A broadly covers solution, emulsion, and suspension routes using cycloaliphatic mercaptans.
- US5807941A discloses a staged modifier strategy in which 10 to 95% of the alkylthiol is charged before polymerization and the remainder is added when conversion reaches 20 to 70%; this is explicitly tied to a low-molecular-weight fraction of 3 to 20 wt.% with Mn not larger than 35,000.
- US3862274A is broader in monomer scope and process scope: it allows polymerizable nitriles such as acrylonitrile or methacrylonitrile, optional vinylidene comonomers, and solution/emulsion/suspension polymerization, with free-radical initiators including azo, peroxide, and redox systems.
- EP2611837B1 shifts the synthesis route from aqueous emulsion to organic-solution polymerization of nitrile rubbers, using at least one conjugated diene, at least one α,β -unsaturated nitrile, an organic solvent, and a molecular-weight modifier selected from mercaptans, mercapto alcohols, mercaptocarboxylic acids/esters, thiocarboxylic acids, disulphides/polysulphides, thiourea, allyl compounds, aldehydes, halohydrocarbons, and saccharin.
- EP2611837B1 is also distinct in explicitly requiring a polar solvent with Hildebrand solubility parameter between 15.5 and 26 (MPa)^{1/2} and allowing the free-radical polymerization to be followed by hydrogenation, linking NBR synthesis to HNBR precursor manufacture.

- US11248078B2 and EP3555198B1 focus less on monomer ratio optimization and more on downstream latex quality: US11248078B2 targets low-salt, color-stable NBR through controlled coagulation and washing, while EP3555198B1 uses post-polymerization thiocarbonyl-functional maturation to improve curability and mechanical properties of latex articles.
- US11248078B2 confirms the conventional industrial baseline for NBR: emulsion polymerization to latex, coagulation with salts or acids, and the importance of residual ion control; it also notes prior art NBR grades with 10 to 30 wt.% acrylonitrile and references staged monomer addition and mercaptan regulation.
- US7592409B2 is not an NBR patent in the strict sense, but it is methodologically relevant because it discloses in-situ emulsifier formation in controlled emulsion polymerization and includes acrylonitrile-containing block copolymer embodiments, showing a newer controlled-polymerization approach rather than classical NBR latex manufacture.
- US7829624B2 is likewise not NBR-specific, but it demonstrates one-batch hydrocarbon-solution synthesis with partial termination and subsequent nanoparticle formation, reinforcing the broader trend toward integrated, multi-step polymerization in a single vessel rather than isolated batch operations.
- US10336853B2 is outside the NBR family, but it is relevant as a process-design comparator: it emphasizes low molecular weight, high T_g, lower initiator and chain-transfer usage, and reduced styrene content, illustrating a modern trend toward minimizing auxiliary reagents while maintaining conversion.
- US9711785B2 is not an NBR synthesis patent; it is included only because it uses styrene-butadiene latex binder systems and therefore provides a downstream latex-application comparator, not a direct low-ACN NBR polymerization teaching.

4. CONCLUSION

The patent evidence supports low-acrylonitrile NBR production primarily through free-radical copolymerization of butadiene with acrylonitrile at the lower end of the conventional NBR composition window, combined with deliberate molecular-weight control and careful latex workup. The most directly applicable synthesis guidance comes from staged-feed and chain-transfer patents: add a substantial portion of the monomer charge initially, then meter the remainder during polymerization, and use mercaptan-type modifiers to suppress high-molecular-weight tails and tune Mooney viscosity. For classical emulsion NBR, the evidence also supports conventional free-radical initiation, with coagulation and washing conditions selected to minimize residual salts and improve color stability and processing reliability. Where solution processing is desired, polar organic solvents and a broad set of molecular-weight modifiers are disclosed, with optional downstream hydrogenation if HNBR precursor flexibility is needed. Overall, the best-supported synthesis strategy for low-ACN NBR is staged emulsion polymerization or polar-solution polymerization with mercaptan-based chain transfer, moderate conversion control, and rigorous coagulation/washing to preserve low-salt, color-stable rubber.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
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- EP2611837B1 | Method for producing nitrile rubbers in organic solvents | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2611837B1/en>
- US7829624B2 | One-pot synthesis of nanoparticles and liquid polymer for rubber applications | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7829624B2/en>
- US3862274A | Preparation of nitrile polymers with cycloaliphatic mercaptans as ... | Not disclosed | US | 1975 | <https://patents.google.com/patent/US3862274A/en>
- US11248078B2 | Color-stable nitrile rubbers | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11248078B2/en>
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- US9711785B2 | Negative electrode slurry composition, lithium ion secondary battery negative ... | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9711785B2/en>
- US20080255314A1 | Dip-forming latex, dip-forming composition and dip-formed article | Not disclosed | US | 2008 | <https://patents.google.com/patent/US20080255314A1/en>
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